

QUANTUM RL AGENT FOR F1 RACE STRATEGY OPTIMIZATION

QC × ML

DIFFICULTY

Advanced

TEAM SIZE

2–4 Members

DURATION

8 Weeks

PRIMARY STACK

PennyLane + PyTorch + OpenAI Gym + FastF1

1. BACKGROUND

Formula 1 race strategy is a highly dynamic optimization problem where decisions related to pit stops, tyre compounds, fuel usage, and race pace adaptation can significantly affect race outcomes. Teams continuously evaluate changing conditions such as tyre degradation, safety car windows, weather changes, traffic conditions, and competitor strategies while making decisions across races that may span more than 70 laps.

The resulting decision space becomes extremely complex due to the large number of interacting variables and possible strategic combinations. Reinforcement Learning has emerged as a powerful approach for solving sequential decision-making problems, while Quantum Machine Learning explores whether variational quantum systems can learn meaningful policies in similarly complex optimization environments.

Quantum Reinforcement Learning (QRL) investigates whether Variational Quantum Circuits (VQCs) can capture relationships between race conditions such as tyre state, lap delta, track conditions, and gap-to-leader information. This project focuses on studying the practical use of QRL models in race strategy optimization while benchmarking them against classical reinforcement learning baselines instead of claiming guaranteed quantum advantage.

2. OBJECTIVE

The objective of this project is to design and train a Quantum Reinforcement Learning agent capable of optimizing Formula 1 race strategies using real historical race telemetry data. Participants will build a custom OpenAI Gym environment and train both a Variational Quantum Circuit (VQC)-based policy agent and a classical DQN baseline under identical simulation conditions.

The project also focuses on benchmarking strategy performance across historical race scenarios while developing replay systems, visualization workflows, and comparative analysis pipelines for evaluating both quantum and classical learning approaches.

Suggested Race Tracks

- Monaco
- Monza
- Silverstone

Note: Teams and circuits above are suggestions only. Participants may choose alternative tracks and teams.

3. SCOPE OF WORK

DATA PIPELINE AND ENVIRONMENT

Participants are expected to build a custom reinforcement learning environment using real Formula 1 race telemetry data. The environment should simulate important race conditions and support reproducible strategy experimentation workflows.

- Build a custom OpenAI Gym environment simulating:

- tyre compound state
- lap delta
- pit stop timing
- safety car periods
- gap-to-leader
- pit loss time

- Use only real race data from the FastF1 API

- No synthetic datasets allowed for benchmarking

- Create a modular and reproducible simulation pipeline

QUANTUM RL AGENT

This phase focuses on implementing a Variational Quantum Circuit-based reinforcement learning agent capable of learning race strategy policies under complex and dynamic race conditions.

- Implement a Variational Quantum Circuit (VQC)-based policy gradient agent

- Use:

- PennyLane
- Qiskit

- Justify:

- qubit count
- gate selection
- entanglement topology
- encoding strategy

- Optimize reward shaping and training stability

- Train the QRL agent on historical race scenarios

CLASSICAL BASELINE

Participants must implement a classical reinforcement learning baseline for benchmarking and comparative evaluation. Both agents should be trained and evaluated under identical environmental conditions.

- Implement a classical DQN baseline
- Train both agents under identical environment conditions
- Compare:
 - convergence speed
 - reward stability
 - sample efficiency
 - race performance

VISUALIZATION AND REPLAY

This phase focuses on developing replay and visualization systems for analyzing race strategies, decision timelines, and comparative agent behavior throughout complete race simulations.

- Build a lap-by-lap race replay system
- Display:
 - QRL decisions
 - DQN decisions
 - real team strategy calls
- Show:
 - resulting position changes
 - tyre changes
 - pit decisions
 - lap delta evolution
- Generate final race summaries and strategy divergence analysis

BENCHMARKING AND ANALYSIS

The final phase focuses on benchmarking the performance of quantum and classical agents while analyzing convergence behavior, training stability, and scalability limitations.

- Measure:
 - cumulative rewards
 - convergence curves
 - training stability
 - race finishing positions
 - strategy divergence
- Compare:
 - QRL agent
 - classical DQN

- real historical team strategies
- Analyze:
 - strengths of QRL approaches
 - limitations of current quantum simulations
 - scalability challenges

4. TECHNICAL CONSTRAINTS

- All race data must be sourced from the FastF1 API
- No synthetic datasets allowed for core benchmarking
- VQC policy network must be implemented using:
 - PennyLane
 - Qiskit
- Use only open-source frameworks and tools
- No proprietary APIs or paid services allowed
- The system must run on consumer-grade hardware
- Classical DQN baseline must be trained under identical environment conditions
- Teams must document:
 - quantum circuit design
 - reward function
 - training methodology

5. DELIVERABLES

MANDATORY

- Trained QRL race strategy agent
- Classical DQN baseline implementation
- Public GitHub repository containing:
 - source code
 - setup instructions
 - training scripts
 - replay system
- Live lap-by-lap replay system showing:
 - agent decisions
 - real team strategy comparison
 - position delta visualization

- Screen-recorded demo video of a full race simulation
- Final technical report containing:
 - reward curves
 - convergence plots
 - sample efficiency analysis
 - strategy divergence analysis
- Final presentation slides summarizing methodology and results

OPTIONAL

- Interactive dashboard for race selection and replay
- Weather-adaptive strategy integration
- Multi-race curriculum learning
- Multi-agent strategy simulation

6. EVALUATION CRITERIA

- Strategy performance compared against:
 - classical DQN baseline
 - historical team strategies
- Technical correctness of:
 - VQC architecture
 - circuit design
 - environment implementation
- Training convergence quality and stability
- Accuracy and clarity of replay visualization
- Code quality, modularity, and reproducibility
- Depth of analysis and report quality

7. SELECTION CRITERIA

- Understanding of:
 - reinforcement learning
 - quantum circuit basics
- Feasibility of implementation plan
- Proposed work distribution across:
 - data engineering

- RL training
 - QML implementation
 - visualization systems
- Prior experience with:
 - PyTorch
 - OpenAI Gym
 - quantum computing frameworks

8. PREREQUISITES

- Python programming
- PyTorch fundamentals
- Reinforcement Learning basics:
 - Q-learning
 - policy gradients
- OpenAI Gym familiarity
- Basic quantum computing concepts
- Enthusiasm for Formula 1 and strategy systems

9. BONUS EXTENSIONS

- Multi-race curriculum learning
- Dynamic weather-adaptive race strategies
- Interactive judge dashboard
- Multi-agent strategy simulation
- Comparative benchmarking across multiple VQC architectures

10. RESOURCES

DATA AND SIMULATION

- FastF1 API
- OpenAI Gym / Gymnasium

QUANTUM MACHINE LEARNING

- PennyLane
- Qiskit
- Qiskit Machine Learning

— [PennyLane QML Tutorials](#)

RESEARCH REFERENCES

— Variational Quantum Circuits for Deep Reinforcement Learning — Skolik et al., 2021

— Quantum agents in the Gym: a variational quantum algorithm for deep Q-learning — Lockwood & Si, 2020

— Playing Atari with Deep Reinforcement Learning — Mnih et al., 2013

MACHINE LEARNING FRAMEWORKS

— [PyTorch](#)